

Lab Problem: Social Network Analysis with R

Problem Statement

You are required to study the **Social Network Analysis with R** project demonstrated in the prescribed video tutorial and implement the demonstrated network-analysis workflow using **R Programming**.

The project should focus on representing relationships among entities as a network, analyzing the structure of the network, identifying important or influential nodes, and visualizing the resulting social network using suitable R packages and functions.

After completing the implementation, execute and test the analysis, interpret the generated network measures and visualizations, and systematically organize the R scripts, dataset, and generated outputs.

Reference Video

Social Network Analysis with R

https://www.youtube.com/watch?v=0xsM0MbRPGE&list=PLpaxqU9O4beceS4RxsJESmFGUqpwm2_P&index=1&t=871s

The supplied video is also referenced as learning material for R-based social/network analytics in academic course material.

Tasks to be Performed

1. Watch the prescribed video and understand the concepts, dataset, network structure, R packages, and implementation procedure demonstrated.
2. Set up the required **R/RStudio environment** and install/load the necessary R packages for social network analysis.
3. Import or prepare the network data required for the analysis.
4. Represent the social network appropriately in R using:
 - **Nodes/vertices** representing entities or individuals.
 - **Edges/links** representing relationships or interactions between them.
5. Implement the social network analysis demonstrated in the prescribed tutorial using appropriate R functions and packages.
6. Generate suitable **network visualizations** to understand the relationships among different nodes.
7. Analyze important characteristics of the network, such as node connections and relevant network measures demonstrated in the video.
8. Identify important, highly connected, or influential nodes based on the results obtained from the analysis.
9. Execute and test the complete R implementation and verify that the generated network measures and visualizations are correct.
10. Provide a brief interpretation of the important observations obtained from the social network analysis.

Deliverables

- Complete R script/source code
- Dataset used for network analysis or appropriate dataset reference
- Network representation generated in R
- Network visualization(s)
- Important network-analysis results/measures
- Brief interpretation of the obtained results
- Screenshot(s) demonstrating successful execution

AI Feedback Scores

Rating: 1 – Lowest | 4 – Highest

Parameter	Score
Real-life Application	4 / 4
Problem-solving Ability	4 / 4
Critical Reasoning	4 / 4
Creativity	3 / 4
Alignment with Learning / Course Outcomes	4 / 4

Estimated Time Required

1 Hour 45 Minutes – 2 Hours

Suggested Time Distribution

- **15–20 Minutes:** Study the prescribed video and understand the concepts of social network analysis and the demonstrated workflow.
- **10–15 Minutes:** Configure R/RStudio, prepare the dataset, and install/load required packages.
- **45–50 Minutes:** Implement the network creation, analysis, and visualization in R.
- **20–25 Minutes:** Execute, test, interpret, and validate the network-analysis results.
- **10 Minutes:** Organize the code, results, screenshots, and final observations.

COs Linked with the Assignment

CO2, CO3 and CO6

CO	Course Outcome	Level of Alignment
CO2	Explain data manipulation techniques (dplyr), visualization (ggplot2), and statistical concepts (hypothesis testing, regression).	Moderate

CO3	Apply R programming to import/export data, perform statistical tests, and implement basic machine learning models.	Strong
CO6	Design and implement an end-to-end data science project using R, from data cleaning to model deployment.	Strong

Primary CO: CO3

LOs Linked with the Assignment

LO 2.1, LO 3.1, LO 3.2, LO 6.1

Suggested Learning Outcomes

- **LO 2.1:** Apply appropriate visualization techniques to represent relationships and structural patterns in network data.
- **LO 3.1:** Apply R programming techniques to import, prepare, represent, and process social network data.
- **LO 3.2:** Implement appropriate R packages and functions for performing social network analysis and calculating relevant network measures.
- **LO 6.1:** Develop an integrated R-based analytical workflow involving data preparation, network construction, analysis, visualization, and interpretation of results.

Justification

This laboratory problem provides students with practical exposure to **Social Network Analysis (SNA)** using R Programming. Instead of analysing observations as independent records, students learn how relationships among entities can be represented using nodes and edges and subsequently analyzed as a network.

The activity has a **high level of real-life application (4/4)** because network-based analysis is widely used to study relationships in social media, communication networks, professional networks, collaboration networks, recommendation systems, organizational structures, citation networks, and several other real-world applications. The supplied video is also referenced in academic materials dealing with R and social/network analytics.

The assignment strongly develops **problem-solving ability (4/4)** because students must understand how the given relationship data should be represented as a network, identify appropriate R functions and libraries, construct the network, generate visualizations, calculate relevant network characteristics, and identify important entities based on the obtained results.

A **critical reasoning score of 4/4** is appropriate because students are required not merely to execute R commands but also to interpret the network structure. For example, they must reason about why particular nodes are more important or better connected, how relationships are distributed across the network, and what the generated network measures indicate about the underlying social structure.

The assignment provides a **moderate-to-high level of creativity (3/4)**. Although the fundamental analysis follows a prescribed tutorial, students can exercise creativity in network visualization, graph layout, node and edge representation, presentation of influential nodes, interpretation of network patterns, and organization of the final analytical output. Since the principal analytical workflow remains guided by the tutorial, a score of 3/4 is more appropriate than 4/4.

The assignment has a **strong alignment with CO3** because students practically apply R programming to work with a specialized form of data, use appropriate R packages, construct analytical objects, execute network-analysis operations, and interpret the resulting output. Therefore, **CO3 is considered the Primary Course Outcome**.

The assignment also supports **CO2**, particularly its visualization component. Network visualization is an important part of Social Network Analysis because graphical representation allows students to examine relationships, connectivity, clusters, and influential entities. However, because the assignment does not necessarily require dplyr, hypothesis testing, or regression, the overall alignment with CO2 should be considered **Moderate rather than Strong**.

CO6 is strongly supported because students undertake an integrated analytical workflow beginning with understanding the problem and preparing the network data, followed by network construction, analysis, visualization, interpretation, testing, and presentation of results. It therefore demonstrates several characteristics of an end-to-end R-based data analytics project.

The activity does **not directly assess CO1, CO4, or CO5**. CO1 primarily concerns fundamental R syntax and basic data structures, which may be used during implementation but are not the principal learning objective. CO4 requires model evaluation techniques such as cross-validation, ROC-AUC, and optimization, which are not inherent to this Social Network Analysis exercise. Similarly, CO5 requires comparison and selection among machine-learning algorithms, which is not part of the prescribed activity. These COs should therefore not be mapped merely to increase CO coverage.

AI Tool Used

ChatGPT

Prompt(s) Used

“Evaluate the given R Programming laboratory problem on **Social Network Analysis with R**, in which students study a prescribed video tutorial and implement the demonstrated social network analysis using R. The activity should include preparation of network data, representation of entities as nodes and relationships as edges, construction and visualization of the network, calculation of relevant network characteristics, identification of important or influential nodes, testing of the implementation, and interpretation of the generated results.

Evaluate the laboratory problem using a **1–4 rubric** for Real-life Application, Problem-solving Ability, Critical Reasoning, Creativity, and Alignment with Learning/Course Outcomes. Estimate the time required and map the assignment specifically against the following R Programming Course Outcomes:

CO1 – fundamental R concepts and data structures;

CO2 – data manipulation, visualization and statistical concepts;

CO3 – application of R for data import/export, statistical tests and basic machine learning;

CO4 – model evaluation and code optimization;

CO5 – comparison and selection of machine learning algorithms; and

CO6 – design and implementation of an end-to-end data science project using R.

Identify only those COs genuinely addressed by the Social Network Analysis activity, identify the Primary CO, provide appropriate Learning Outcomes, and give detailed academic justification for the scores and CO mapping.”